SAI SHARMILA KINJARAPU

INTEGRATED M.TECH | SOFTWARE ENGINEERING | APPLIED AI/ML | DATA ANALYTICS

Amaravati, Andhra Pradesh, India | +91 9059564121 | saisharmilak2005@gmail.com | LinkedIn | GitHub

SUMMARY

Fourth-year Integrated M.Tech Software Engineering student at VIT-AP (expected 2028; CGPA 9.13/10) with project experience across satellite-image segmentation, a multi-agent GenAI prototype and Flask/MongoDB application development. Completed Google training in data analytics, advanced analytics and business intelligence; interested in software engineering, applied AI/ML and data-driven systems.

EDUCATION

VIT-AP University | Integrated M.Tech in Software Engineering | 2023–2028 (Expected)

CGPA: 9.13/10 | Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Networks, Machine Learning, Data Science and Software Testing (coursework).

TECHNICAL SKILLS

Programming & Querying: Python, Java, SQL, R

Machine Learning & Vision: PyTorch, CNNs, U-Net, ResNet34 U-Net, YOLO11-seg, transfer learning, segmentation metrics, Grad-CAM

Applied GenAI: Gemini, Google ADK, multi-agent workflows, prompt engineering, LLM fundamentals

Analytics & Data: Data cleaning, exploratory analysis, regression/predictive modeling, Tableau, spreadsheets, BigQuery coursework

Development & Libraries: Flask, MongoDB, HTML/CSS, Git, VS Code, Gradio, NumPy, pandas, OpenCV, scikit-learn, h5py, SciPy, xarray/NetCDF

PROJECTS & APPLIED WORK

TerraSeg AI - Multi-Dataset Satellite Landslide Segmentation Research Prototype (In Progress) | *Python, PyTorch, ResNet34 U-Net, YOLO11-seg, OpenCV, scikit-learn, Grad-CAM, Gradio*

GitHub repository

- Implemented experimental segmentation workflows across 3,799 Landslide4Sense 14-channel patches (12 optical bands, slope and DEM) and 770 Bijie landslide image-mask pairs; handled preprocessing, channel normalisation, data splits and model evaluation.
- Compared Custom CNN, U-Net/ResNet34 U-Net, YOLO11-seg and Random Forest; tested BCE/Dice and Focal/Dice losses, augmentation, SCSE/C2PSA attention and alternative channel combinations.
- Three-seed results included 0.6079 mean test IoU / 0.7562 F1 for the All14 U-Net setup on Landslide4Sense, and 0.7913 mean test IoU / 0.8835 F1 for ResNet34 U-Net on the selected Bijie landslide subset.
- Implemented Grad-CAM, channel-permutation analysis, connected-component candidate-region extraction (boxes, polygons and JSON) and a Gradio demo for patch inspection; consolidating the research workflow into a more reproducible interactive prototype remains in progress.

Evaluation caveat: the reported Bijie subset contains landslide-positive samples only; results do not establish false-alarm performance on non-landslide imagery, operational readiness or geospatial accuracy.

AquaGuard AI - Multi-Agent Aquaculture Decision-Support Prototype | *Python, Gemini, Google ADK*

GitHub repository

- Built a four-agent prototype with a coordinator and specialist water-quality, weather and market agents; the water analysis considers dissolved oxygen, pH, temperature and salinity, while the other agents structure weather and market context.
- Synthesizes user-supplied aquaculture information into evidence, suggested next steps and uncertainty. Uses supplied/demo inputs only; live IoT, external weather/market feeds and Firebase/Firestore integrations are not implemented.

Hospital Management System - Software Learning Prototype | *Python, Flask, MongoDB, HTML/CSS, Git*

GitHub repository

- Worked on components of a role-based Flask/MongoDB application scaffold; the repository documents role-specific dashboards and workflows for appointments, prescriptions, billing, reports, users and schedules.
- Used AI tools as a learning aid to explore prompt design and how instructions/context influence LLM responses while reviewing and iterating on implementation; this is a learning prototype, not a production deployment.

AI Planning & Search Coursework | *Python, Java, A*, Minimax, BFS, PDDL*

GitHub repository

- Coursework includes an A* 8-puzzle solver, Minimax Tic-Tac-Toe and a BFS water-jug problem; additional planning work includes a Python travel planner using goal-stack planning and PDDL.

CERTIFICATIONS & ANALYTICS TRAINING

Google Data Analytics Professional Certificate; Google Advanced Data Analytics Professional Certificate; Google Business Intelligence Certificate; Prompt Engineering for ChatGPT - Vanderbilt University (Coursera).

Analytics training: SQL/BigQuery, Python, R, data cleaning, exploratory analysis, regression/predictive modeling, spreadsheets and Tableau visualization.

LEADERSHIP & HIGHLIGHTS

- Student Lead, Computer Society of India (CSI), VIT-AP - supported peer-learning, technical awareness and student-engagement initiatives.
- Participated in Gen AI Impact Edition (APAC).